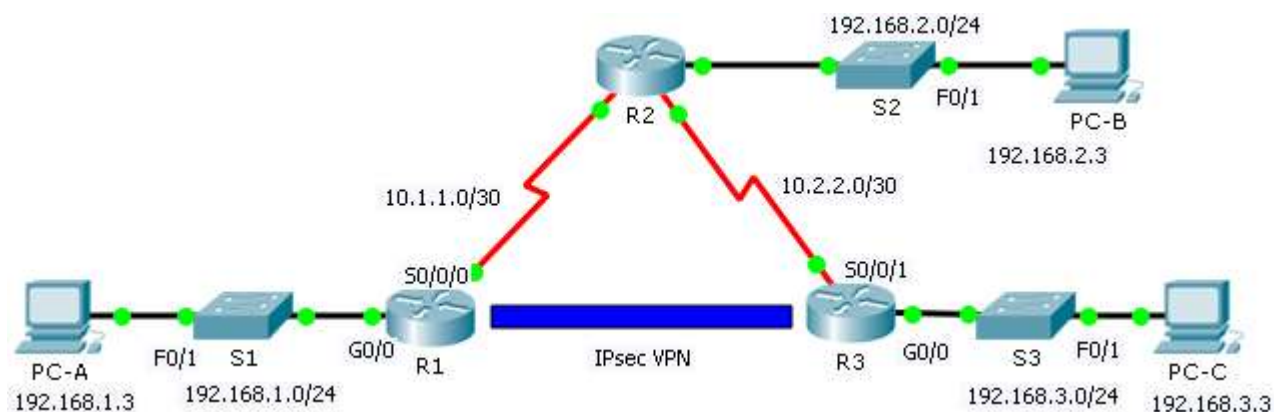


Configuring Site-to-Site VPNs

Introduction

In this lab, I learned how to configure a site-to-site IPsec VPN using Cisco Packet Tracer to securely connect two separate LANs over an untrusted network. The lab focused on enabling security features on Cisco routers, creating access control lists to identify interesting traffic, and configuring ISAKMP Phase 1 and IPsec Phase 2 settings. I also explored how to apply crypto maps and verify tunnel establishment through packet encryption and decryption, simulating secure communication between two remote networks.

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	10.1.1.2	255.255.255.252	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A
	S0/0/0	10.1.1.1	255.255.255.252	N/A
	S0/0/1	10.2.2.1	255.255.255.252	N/A
R3	G0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/1	10.2.2.2	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1

PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1
------	-----	-------------	---------------	-------------

ISAKMP Phase 1 Policy Parameters

Parameters		R1	R3
Key distribution method	Manual or ISAKMP	ISAKMP	ISAKMP
Encryption algorithm	DES , 3DES, or AES	AES	AES
Hash algorithm	MD5 or SHA-1	SHA-1	SHA-1
Authentication method	Pre-shared keys or RSA	pre-share	pre-share
Key exchange	DH Group 1, 2, or 5	DH 2	DH 2
IKE SA Lifetime	86400 seconds or less	86400	86400
ISAKMP Key		cisco	cisco

Bolded parameters are defaults. Other parameters need to be explicitly configured.

IPsec Phase 2 Policy Parameters

Parameters	R1	R3
Transform Set	VPN-SET	VPN-SET
Peer Hostname	R3	R1
Peer IP Address	10.2.2.2	10.1.1.2
Network to be encrypted	192.168.1.0/24	192.168.3.0/24
Crypto Map name	VPN-MAP	VPN-MAP
SA Establishment	ipsec-isakmp	ipsec-isakmp

Objectives

Part 1: Enable Security Features

Part 2: Configure IPsec Parameters on R1

Part 3: Configure IPsec Parameters on R3

Part 4: Verify the IPsec VPN

Scenario

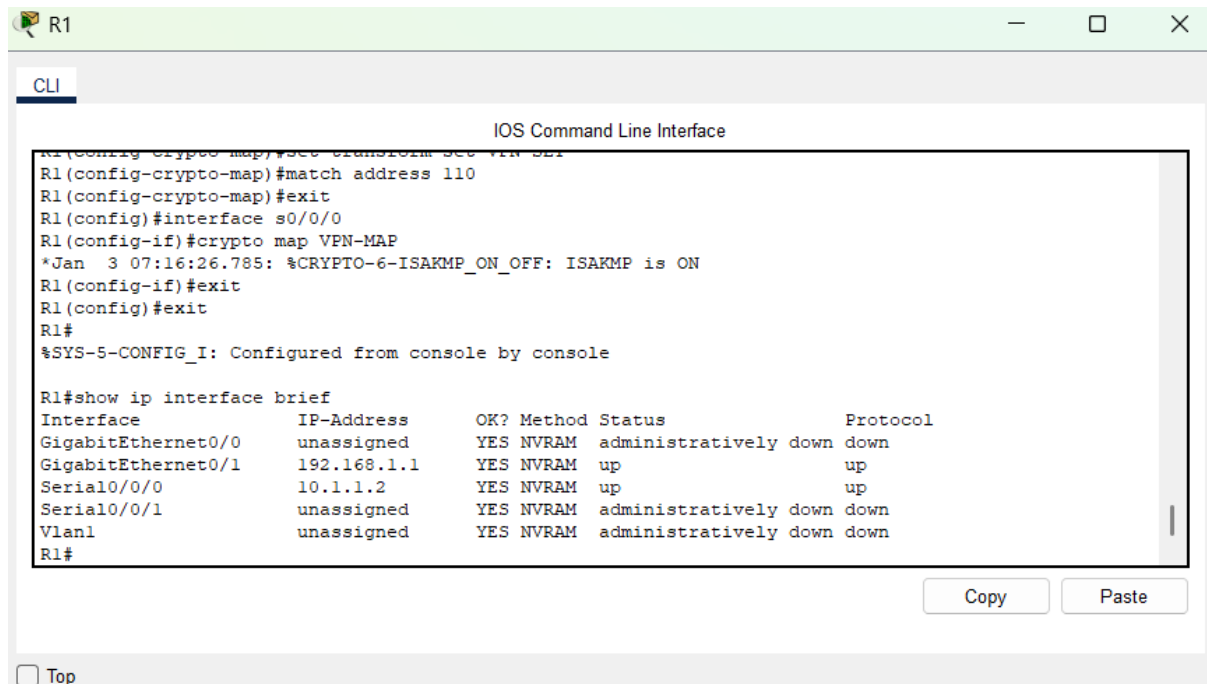
In this activity, I will configure two routers to support a site-to-site IPsec VPN for traffic flowing from their respective LANs. The IPsec VPN traffic will pass through another router that has no knowledge of the VPN. IPsec provides secure transmission of sensitive information over unprotected networks such as the Internet. IPsec acts at the network layer,

protecting and authenticating IP packets between participating IPsec devices (peers), such as Cisco routers

Part 1: Enable Security Features

In this section, I learned how to enable security features on Cisco routers using the securityk9 module.

Confirming configurations:



The screenshot shows a Cisco IOS CLI window titled "R1" with a "CLI" tab. The window displays the following commands and output:

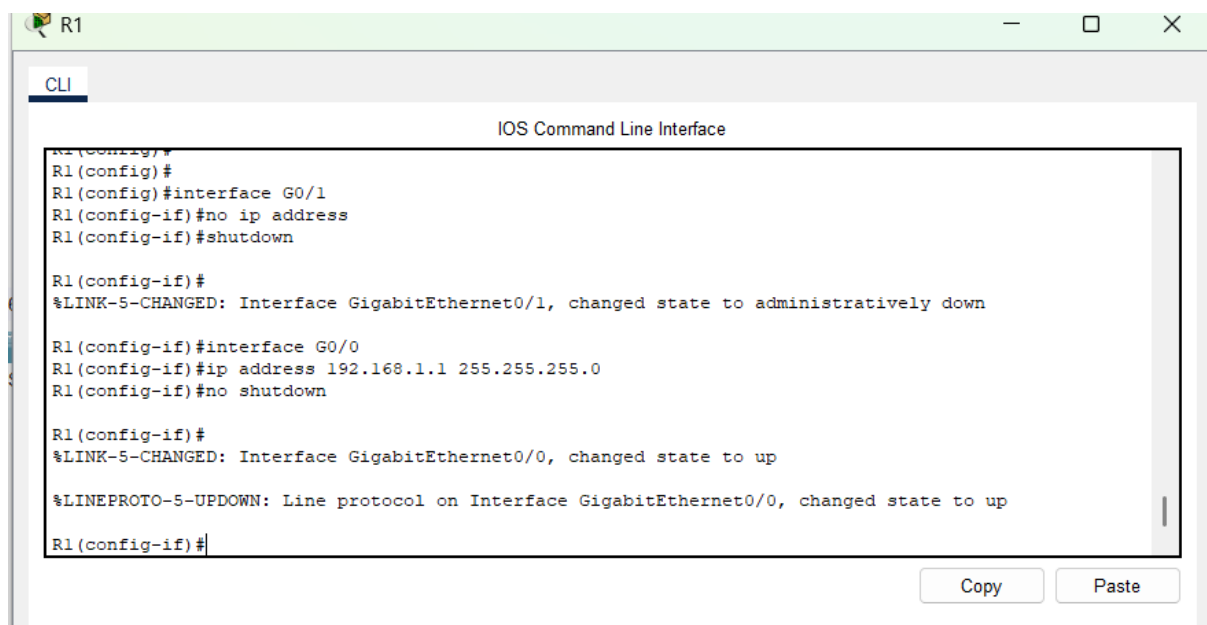
```
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#interface s0/0/0
R1(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/1	192.168.1.1	YES	NVRAM	up	up
Serial0/0/0	10.1.1.2	YES	NVRAM	up	up
Serial0/0/1	unassigned	YES	NVRAM	administratively down	down
Vlan1	unassigned	YES	NVRAM	administratively down	down

At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" link.

Noting that R1 configurations were not according to the addressing table I disconnected the cable from G0/1 to G0/0 then ran the following commands to correct the mismatch



The screenshot shows a Cisco IOS CLI window titled "R1" with a "CLI" tab. The window displays the following commands and output:

```
R1(config)#
R1(config)#interface G0/1
R1(config-if)#no ip address
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

R1(config-if)#interface G0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

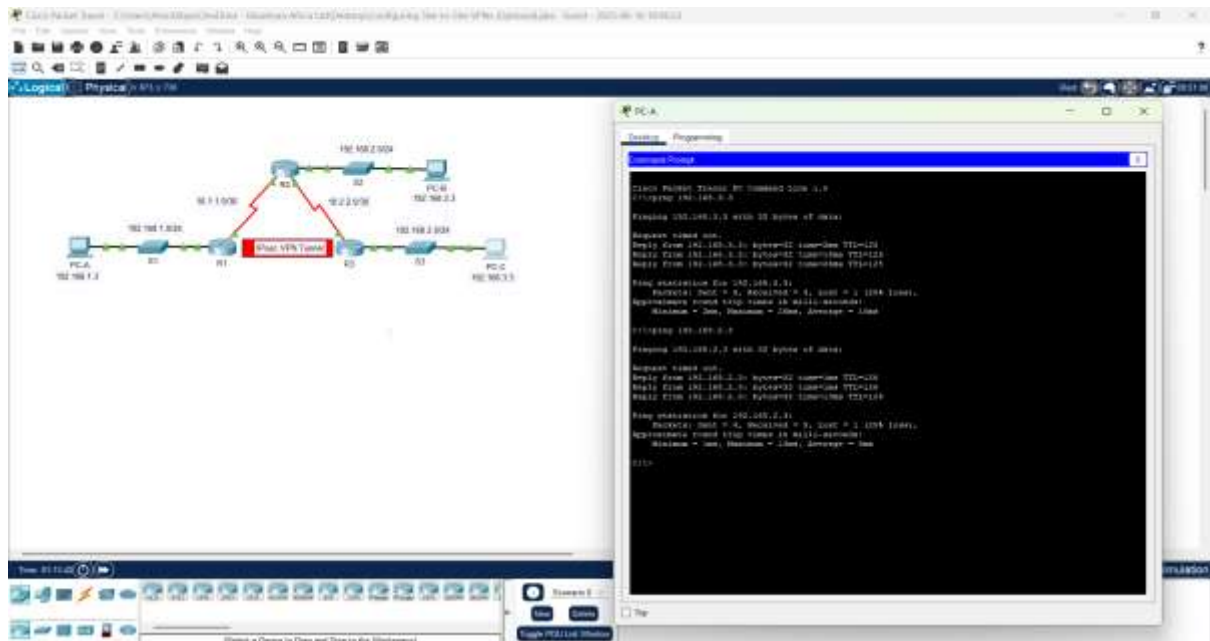
R1(config-if)#
```

At the bottom of the window, there are "Copy" and "Paste" buttons.

The commands start by removing the ip addressing from G0/1 and shutting it down then assigning the ip addressing to the correct interface which according to the table is G0/0.

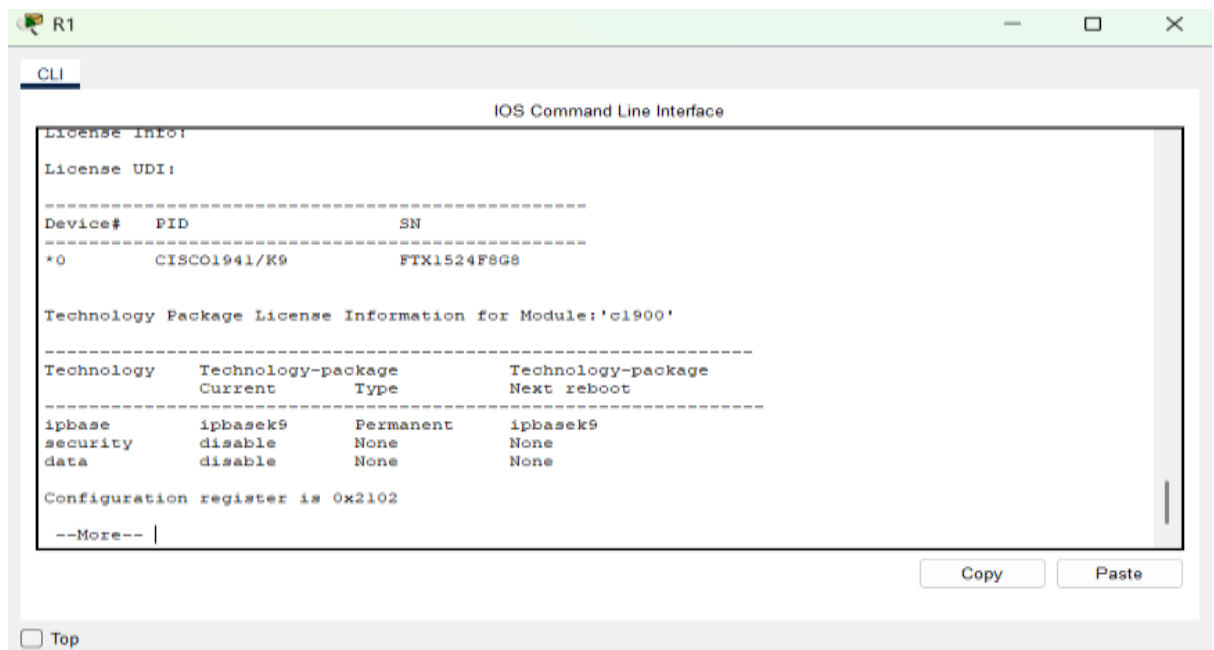
Step1: Test Connectivity

I started by testing connectivity where from PCA pinged PCB and PCC which was a success.



Step 2: Enable the Security Technology Package

1. **Verified the license status** using the show version command to check if the security technology package was active.



```
R2
CLI
IOS Command Line Interface

License UDI:
-----
Device# PID SN
-----
*0 CISC01941/K9 FTX1524RRQ4

Technology Package License Information for Module:'c1900'
-----
Technology Technology-package Technology-package
Current Type Next reboot
-----
ipbase ipbasek9 Permanent ipbasek9
security disable None None
data disable None None

Configuration register is 0x2102

--More--
Copy Paste
Top
```

```
R3
CLI
IOS Command Line Interface

Device# PID SN
-----
*0 CISC01941/K9 FTX1524I27D

Technology Package License Information for Module:'c1900'
-----
Technology Technology-package Technology-package
Current Type Next reboot
-----
ipbase ipbasek9 Permanent ipbasek9
security securityk9 Evaluation securityk9
data disable None None

Configuration register is 0x2102

R3#
R3#
Copy Paste
Top
```

From the 3 screenshots, only one R3 has the security technology active. Hence the next step is to activate the technology in R1 and R2.

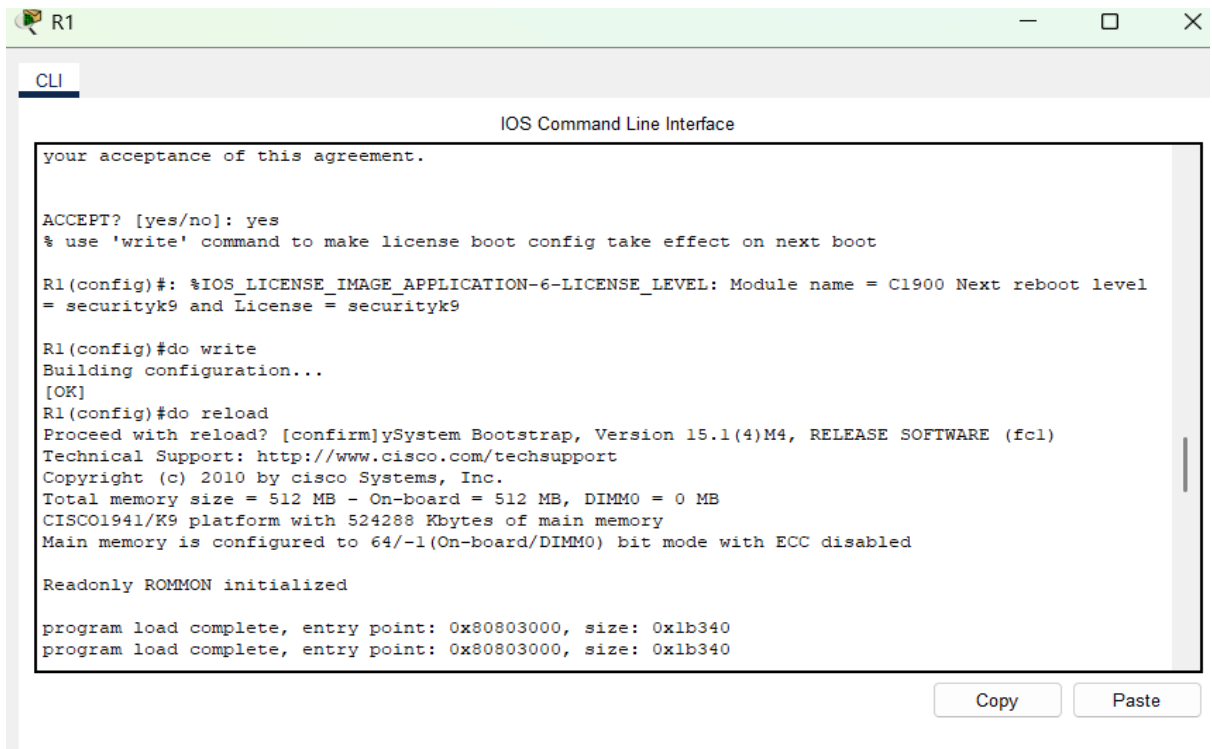
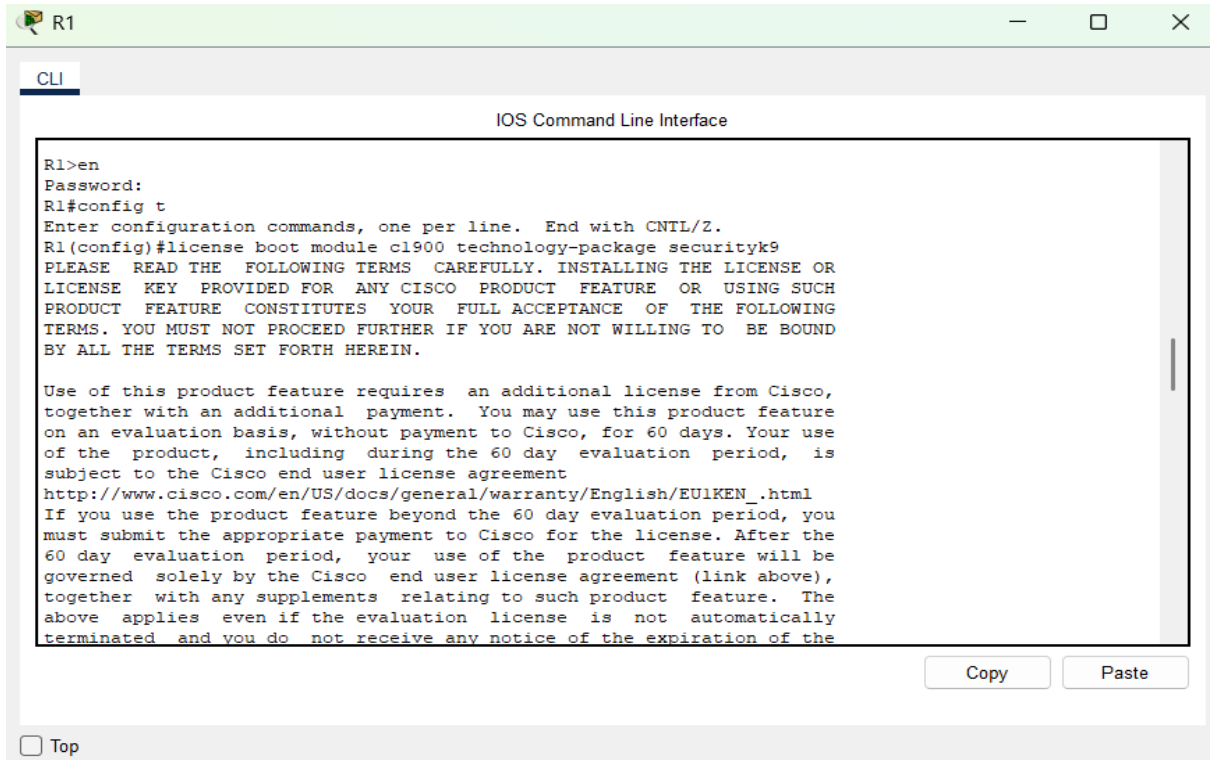
2. **Activated the securityk9 module** if it was not enabled:

R1(config)# license boot module c1900 technology-package securityk9

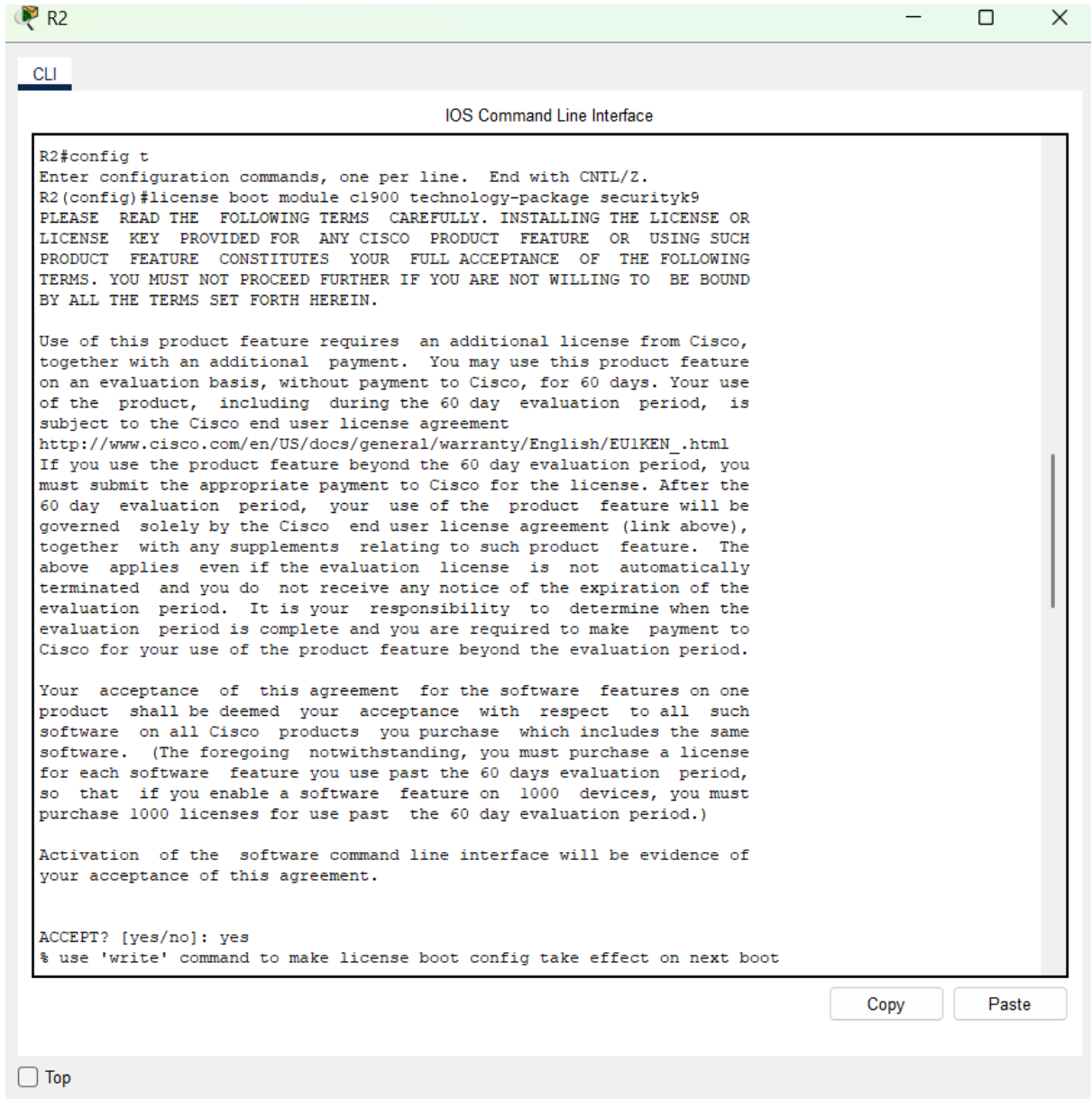
R1(config)# end

R1# do write

R1# reload



3. Repeated the above steps on router R2.



```
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#license boot module cl900 technology-package securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING SUCH
PRODUCT FEATURE CONSTITUTES YOUR FULL ACCEPTANCE OF THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires an additional license from Cisco,
together with an additional payment. You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the product, including during the 60 day evaluation period, is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EULKEN.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day evaluation period, your use of the product feature will be
governed solely by the Cisco end user license agreement (link above),
together with any supplements relating to such product feature. The
above applies even if the evaluation license is not automatically
terminated and you do not receive any notice of the expiration of the
evaluation period. It is your responsibility to determine when the
evaluation period is complete and you are required to make payment to
Cisco for your use of the product feature beyond the evaluation period.

Your acceptance of this agreement for the software features on one
product shall be deemed your acceptance with respect to all such
software on all Cisco products you purchase which includes the same
software. (The foregoing notwithstanding, you must purchase a license
for each software feature you use past the 60 days evaluation period,
so that if you enable a software feature on 1000 devices, you must
purchase 1000 licenses for use past the 60 day evaluation period.)

Activation of the software command line interface will be evidence of
your acceptance of this agreement.

ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot
```

☐ Top

```
IOS Command Line Interface

= securityk9 and License = securityk9

R2(config)#do write
Building configuration...
[OK]
R2(config)#do reload
Proceed with reload? [confirm]ySystem Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1 (On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem
      TYPE      MEMORY REQ
      HWIC Slot 0      0x00200000
      HWIC Slot 1      0x00200000      Onboard devices &
      buffer pools      0x01E8F000
-----
      TOTAL:      0x02E8F000
Rounded IOMEM up to: 48Mb.
Using 6 percent iomem. [48Mb/512Mb]

      Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
```

Copy Paste

4. **After rebooting**, I confirmed activation using show version again and R1 and R2 were now activated.

R1

CLI

IOS Command Line Interface

249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

Device#	PID	SN
*0	CISCO1941/K9	FTX1524F8G8

Technology Package License Information for Module:'c1900'

Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	securityk9	Evaluation	securityk9
data	disable	None	None

Configuration register is 0x2102

--More--

Copy Paste

☐ Top

R2

CLI

IOS Command Line Interface

249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

Device#	PID	SN
*0	CISCO1941/K9	FTX1524RRQ4

Technology Package License Information for Module:'c1900'

Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	securityk9	Evaluation	securityk9
data	disable	None	None

Configuration register is 0x2102

R2#

Copy Paste

In this section,

- The command `show version` is used to verify whether the security features (like IPsec) are enabled on the router. This is needed for VPN configuration.

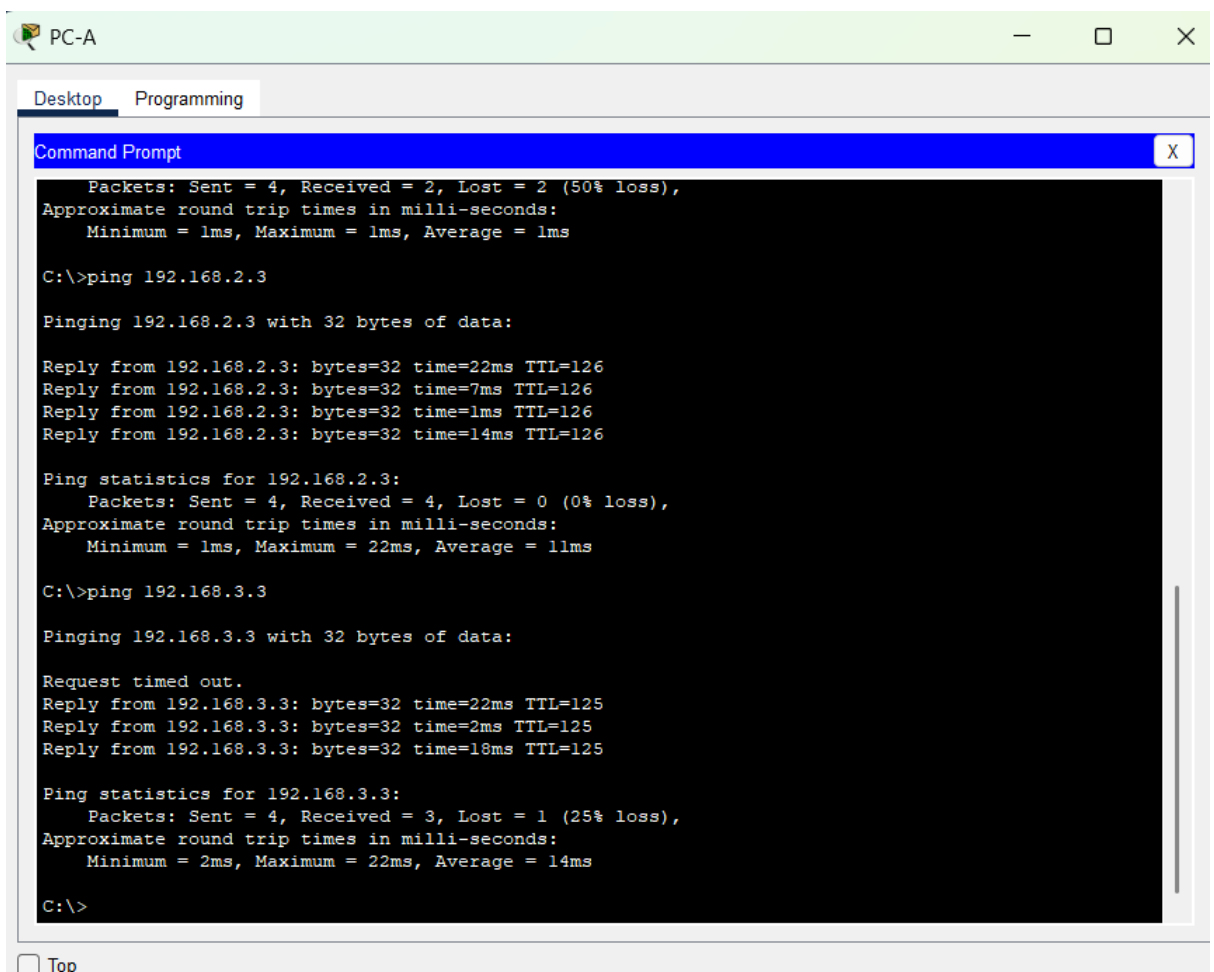
- If security features are not active, the command `license boot module c1900 technology-package securityk9` enables them. This command tells the router to boot with the required security package.
- After making changes, the command `copy running-config startup-config` or `do write` saves the configuration so it's not lost after a reboot.
- The `reload` command restarts the router to activate the new license.

Part 2: Configure IPsec Parameters on R1

In this section, I learned how to configure IPsec VPN settings on R1 using access control lists, ISAKMP, and IPsec crypto maps.

Steps followed:

1. I Pinged PC-C from PC-A to verify basic network connectivity.



```

PC-A
Desktop Programming
Command Prompt
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=22ms TTL=126
Reply from 192.168.2.3: bytes=32 time=7ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 22ms, Average = 11ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

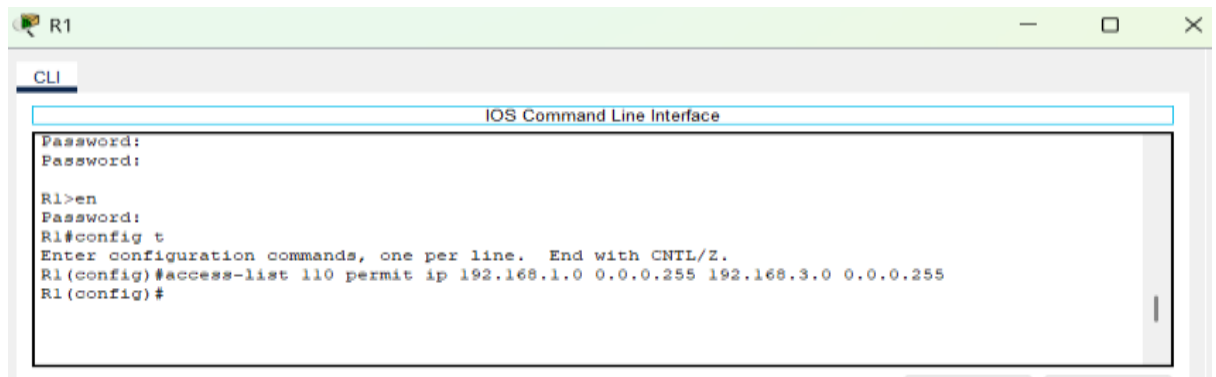
Request timed out.
Reply from 192.168.3.3: bytes=32 time=22ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
Reply from 192.168.3.3: bytes=32 time=18ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 22ms, Average = 14ms

C:\>
  
```

2. Created an ACL (110) to define interesting traffic (LAN-to-LAN):

```
R1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```



```
R1
CLI
IOS Command Line Interface
Password:
Password:
R1>en
Password:
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
R1(config)#
```

3. Configured ISAKMP Phase 1 on R1:

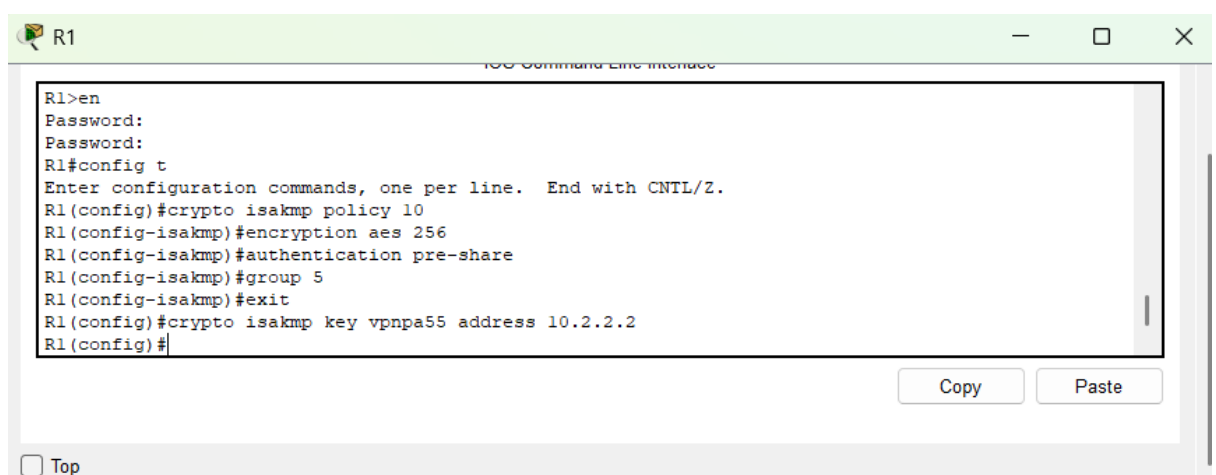
```
R1(config)# crypto isakmp policy 10
```

```
R1(config-isakmp)# encryption aes
```

```
R1(config-isakmp)# authentication pre-share
```

```
R1(config-isakmp)# group 2
```

```
R1(config)# crypto isakmp key cisco address 10.2.2.2
```



```
R1
IOS Command Line Interface
R1>en
Password:
Password:
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key vpnpa55 address 10.2.2.2
R1(config)#
```

Copy Paste

☐ Top

4. Configured IPsec Phase 2 (transform set and crypto map):

```
R1(config)# crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac
```

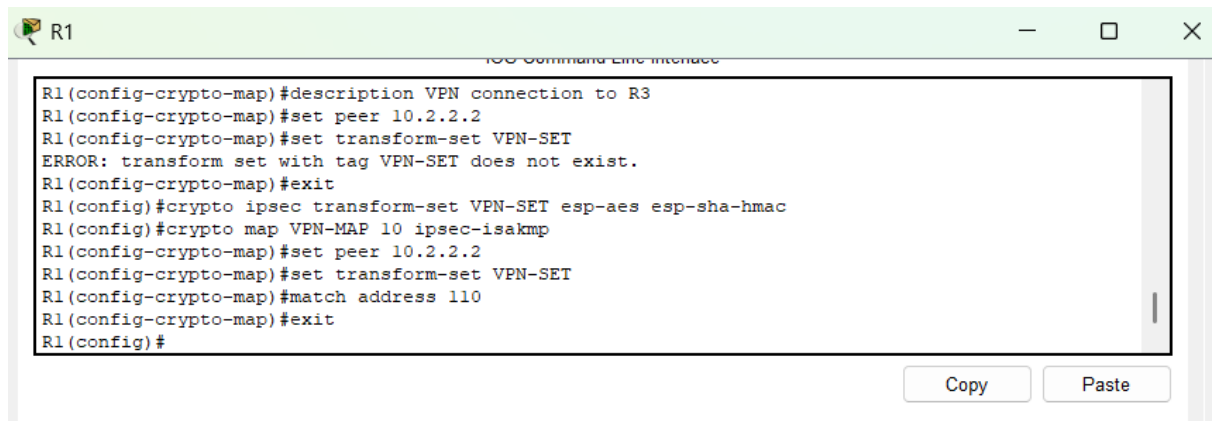
```
R1(config)# crypto map VPN-MAP 10 ipsec-isakmp
```

```
R1(config-crypto-map)# description VPN connection to R3
```

```
R1(config-crypto-map)# set peer 10.2.2.2
```

```
R1(config-crypto-map)# set transform-set VPN-SET
```

```
R1(config-crypto-map)# match address 110
```

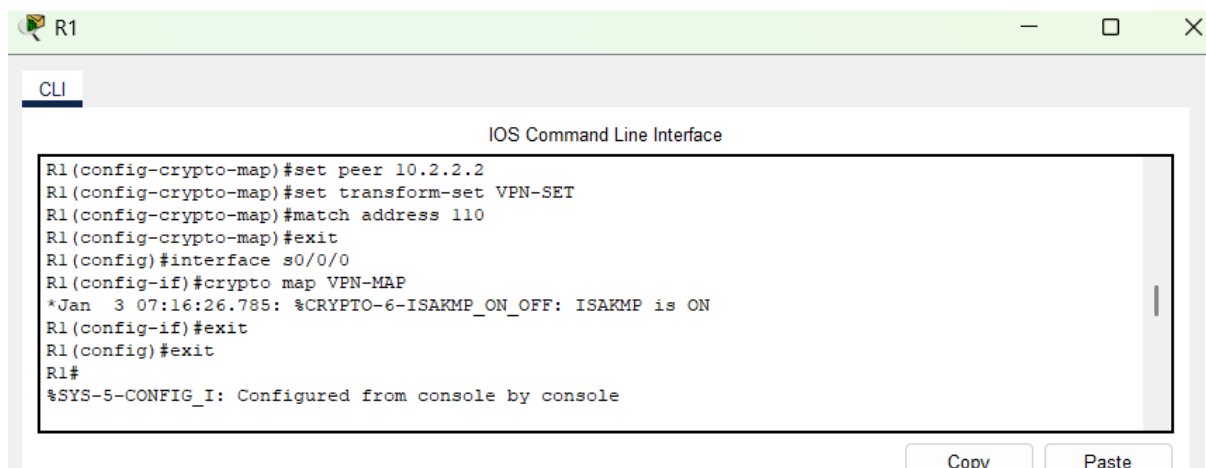
A screenshot of a terminal window titled "R1" with a green header bar. The terminal shows the configuration of a crypto map on R1. The commands entered are: `R1(config-crypto-map)#description VPN connection to R3`, `R1(config-crypto-map)#set peer 10.2.2.2`, `R1(config-crypto-map)#set transform-set VPN-SET`, which results in an error: `ERROR: transform set with tag VPN-SET does not exist.`, `R1(config-crypto-map)#exit`, `R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac`, `R1(config)#crypto map VPN-MAP 10 ipsec-isakmp`, `R1(config-crypto-map)#set peer 10.2.2.2`, `R1(config-crypto-map)#set transform-set VPN-SET`, `R1(config-crypto-map)#match address 110`, `R1(config-crypto-map)#exit`, and `R1(config)#`. At the bottom right, there are "Copy" and "Paste" buttons.

```
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
ERROR: transform set with tag VPN-SET does not exist.
R1(config-crypto-map)#exit
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#
```

5. I Applied the crypto map to the outgoing interface:

`R1(config)# interface s0/0/0`

`R1(config-if)# crypto map VPN-MAP`

A screenshot of a terminal window titled "R1" with a green header bar. The terminal shows the application of the crypto map to the interface. The commands entered are: `R1(config-crypto-map)#set peer 10.2.2.2`, `R1(config-crypto-map)#set transform-set VPN-SET`, `R1(config-crypto-map)#match address 110`, `R1(config-crypto-map)#exit`, `R1(config)#interface s0/0/0`, `R1(config-if)#crypto map VPN-MAP`, which results in a message: `*Jan 3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON`, `R1(config-if)#exit`, `R1(config)#exit`, and `R1#`. At the bottom, a system message is displayed: `%SYS-5-CONFIG_I: Configured from console by console`. At the bottom right, there are "Copy" and "Paste" buttons.

```
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#interface s0/0/0
R1(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

In summary in this section I did the following:

Step 1: Define Interesting Traffic

- The command `access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255` defines which traffic should be encrypted over the VPN. This is called "interesting traffic"—in this case, traffic from R1's LAN to R3's LAN.

Step 2: ISAKMP Phase 1 Configuration

- `crypto isakmp policy 10` creates a new IKE policy and assigns it a priority of 10. An IKE policy (Internet Key Exchange policy) is a set of rules that define how two VPN peers (like routers or firewalls) establish a secure tunnel between them—specifically for Phase 1 of the IPsec VPN negotiation.
- `encryption aes` sets the encryption algorithm to AES for securing Phase 1 traffic.
- `authentication pre-share` configures the use of a pre-shared key for authentication.

- `group 2` sets the Diffie-Hellman group for key exchange to group 2, which provides a balance of security and speed.
- `crypto isakmp key cisco address 10.2.2.2` sets the shared secret key (in this case, "cisco") and defines the IP address of the remote peer (R3).

Step 3: Configure IPsec Phase 2 (Transform Set and Crypto Map)

- `crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac` defines the encryption (3DES) and hashing (SHA) algorithms for encrypting the data.
- `crypto map VPN-MAP 10 ipsec-isakmp` starts the creation of a crypto map and links it to ISAKMP.
- `set peer 10.2.2.2` tells the router who the VPN peer is (R3's serial IP).
- `set transform-set VPN-SET` links the earlier encryption settings to this VPN.
- `match address 110` binds the access list (the defined interesting traffic) to the crypto map.

Step 4: Apply the Crypto Map to the Interface

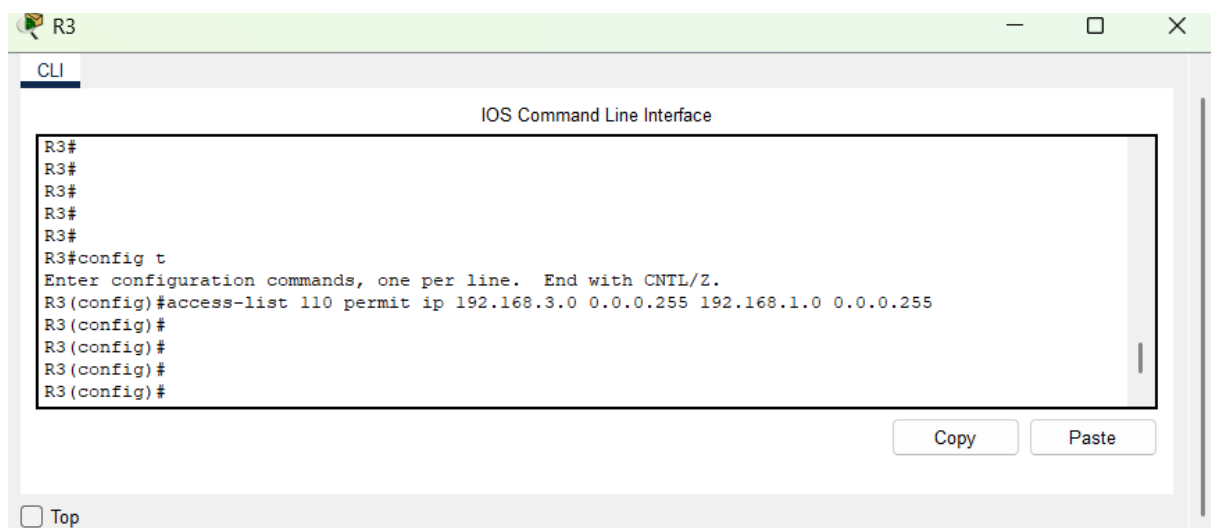
- `interface s0/0/0` enters the serial interface configuration mode.
- `crypto map VPN-MAP` applies the VPN configuration to this physical interface so the router knows to secure the traffic that passes through it.

Part 3: Configure IPsec Parameters on R3

In this section, I learned to configure the same IPsec VPN settings on R3 to establish a site-to-site tunnel with R1.

1. **Created ACL (110)** for interesting traffic from R3 to R1:

```
R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```



2. **Configured ISAKMP Phase 1 on R3:**

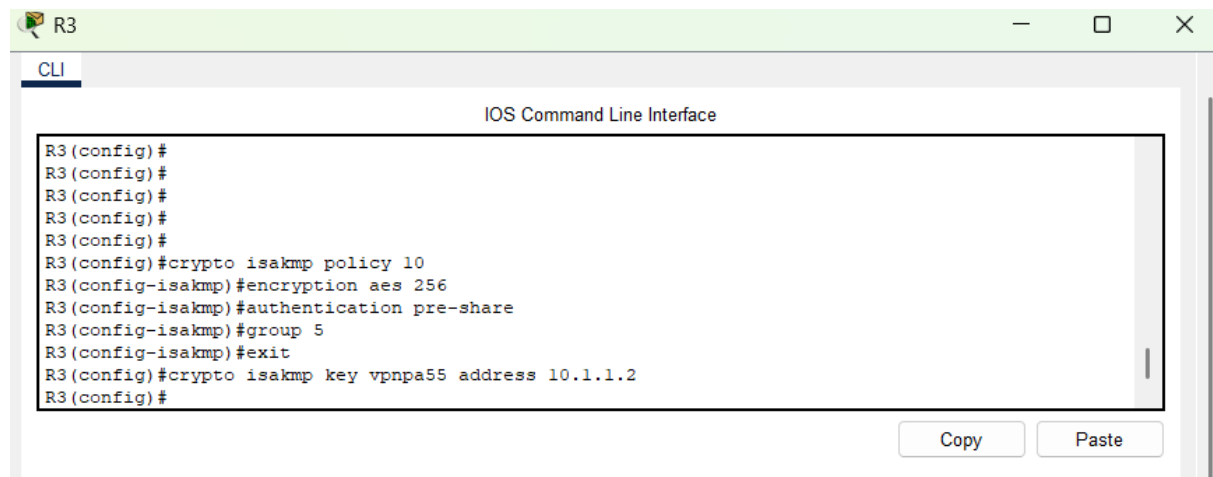
R3(config)# crypto isakmp policy 10

R3(config-isakmp)# encryption aes

R3(config-isakmp)# authentication pre-share

R3(config-isakmp)# group 2

R3(config)# crypto isakmp key cisco address 10.1.1.2



A screenshot of a network simulator window titled 'R3'. The window shows the 'CLI' (Command Line Interface) tab. The text area displays the following commands and their prompts:

```
R3(config)#  
R3(config)#  
R3(config)#  
R3(config)#  
R3(config)#  
R3(config)#crypto isakmp policy 10  
R3(config-isakmp)#encryption aes 256  
R3(config-isakmp)#authentication pre-share  
R3(config-isakmp)#group 5  
R3(config-isakmp)#exit  
R3(config)#crypto isakmp key vpnpa55 address 10.1.1.2  
R3(config)#
```

At the bottom right of the text area, there are two buttons: 'Copy' and 'Paste'.

3. Configured IPsec Phase 2 (transform set and crypto map):

R3(config)# crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac

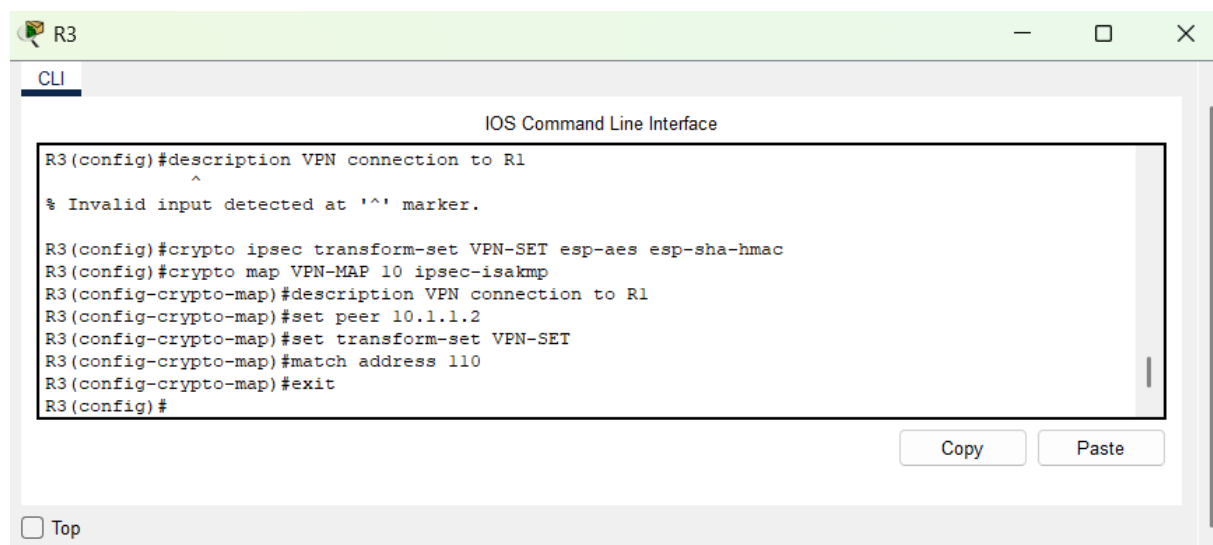
R3(config)# crypto map VPN-MAP 10 ipsec-isakmp

R3(config-crypto-map)# description VPN connection to R1

R3(config-crypto-map)# set peer 10.1.1.2

R3(config-crypto-map)# set transform-set VPN-SET

R3(config-crypto-map)# match address 110



A screenshot of a network simulator window titled 'R3'. The window shows the 'CLI' (Command Line Interface) tab. The text area displays the following commands and their prompts:

```
R3(config)#description VPN connection to R1  
^  
% Invalid input detected at '^' marker.  
  
R3(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac  
R3(config)#crypto map VPN-MAP 10 ipsec-isakmp  
R3(config-crypto-map)#description VPN connection to R1  
R3(config-crypto-map)#set peer 10.1.1.2  
R3(config-crypto-map)#set transform-set VPN-SET  
R3(config-crypto-map)#match address 110  
R3(config-crypto-map)#exit  
R3(config)#
```

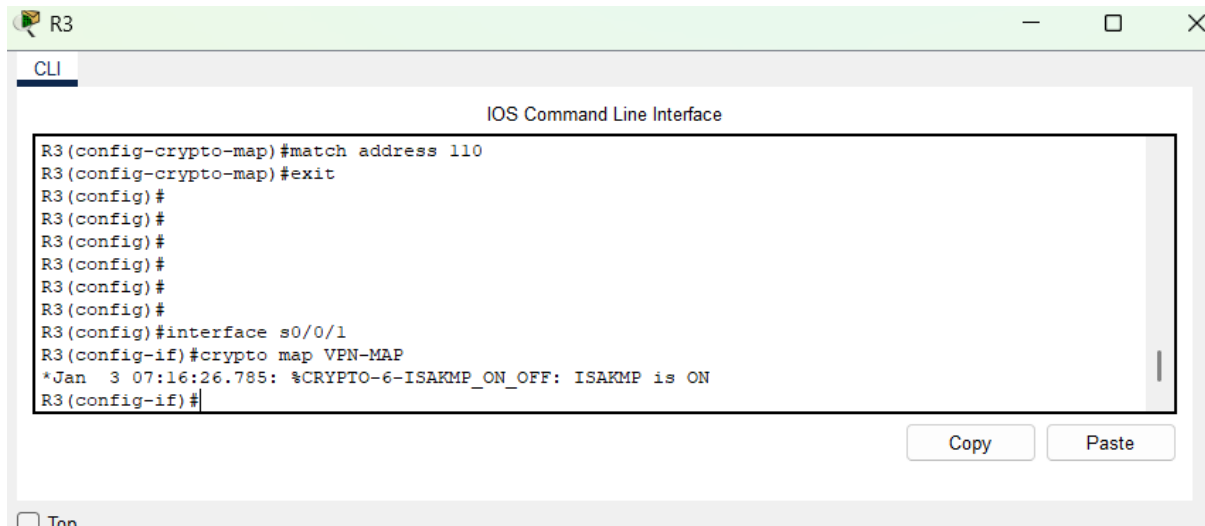
At the bottom right of the text area, there are two buttons: 'Copy' and 'Paste'.

At the bottom left of the window, there is a checkbox labeled 'Top'.

4. Applied the crypto map to outgoing interface s0/0/1:

```
R3(config)# interface s0/0/1
```

```
R3(config-if)# crypto map VPN-MAP
```



```
R3
CLI
IOS Command Line Interface

R3(config-crypto-map)#match address 110
R3(config-crypto-map)#exit
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#interface s0/0/1
R3(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R3(config-if)#
```

In Summary, this section covered the following:

Step 1: Define Interesting Traffic

- `access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255` marks the traffic coming from R3's LAN and going to R1's LAN as interesting.

Step 2: ISAKMP Phase 1 Configuration

- `crypto isakmp policy 10` creates a matching IKE policy.
- `encryption aes` sets AES as the encryption algorithm.
- `authentication pre-share` configures the same shared key method.
- `group 2` sets the key exchange method.
- `crypto isakmp key cisco address 10.1.1.2` specifies the same pre-shared key and points to the R1 peer IP.

Step 3: IPsec Phase 2 Configuration

- `crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac` defines the encryption and integrity mechanisms.
- `crypto map VPN-MAP 10 ipsec-isakmp` creates the crypto map container.
- `set peer 10.1.1.2` sets the VPN peer (R1's IP).
- `set transform-set VPN-SET` connects the crypto map to the transform set.
- `match address 110` binds the ACL to the VPN.

Step 4: Apply to Interface

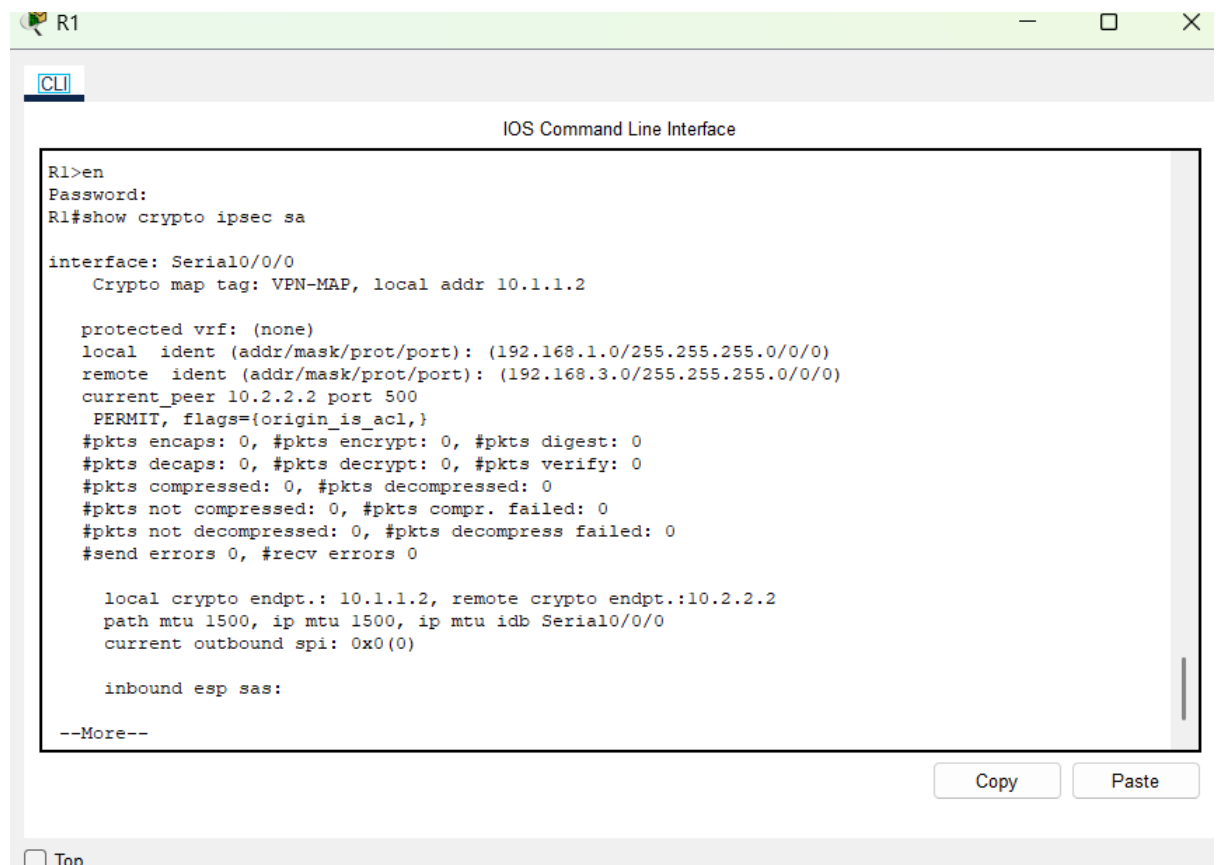
- `interface s0/0/1` accesses R3's serial interface.
- `crypto map VPN-MAP` activates VPN settings on that interface.

Part 4: Verify the IPsec VPN

In this section, I learned how to verify VPN tunnel establishment and observe packet encryption behavior.

1. Checked IPsec status before traffic:

R1# show crypto ipsec sa



```
R1>en
Password:
R1#show crypto ipsec sa

interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

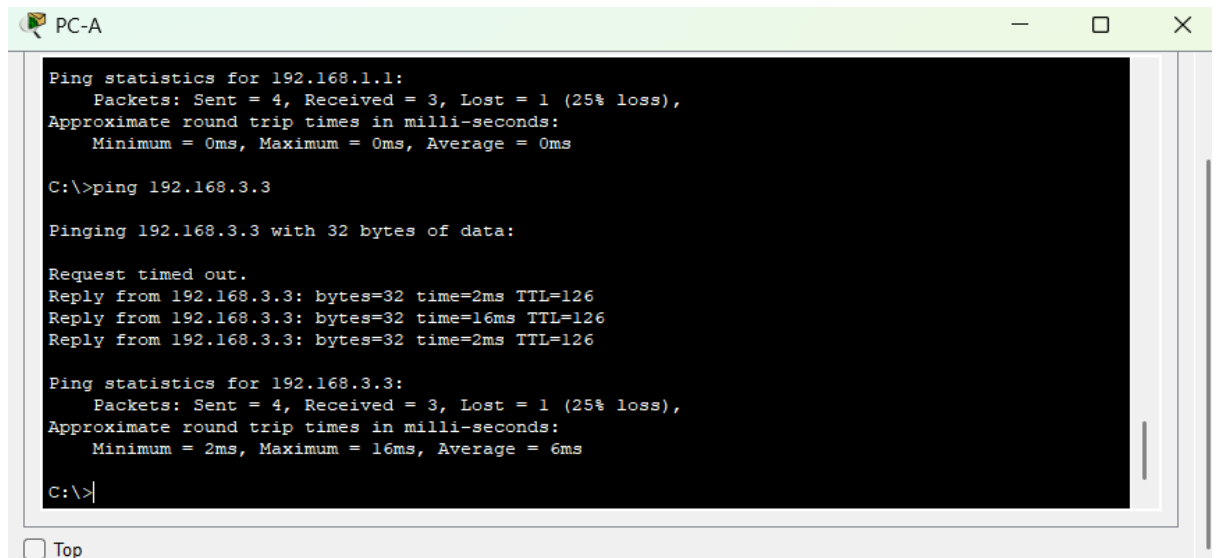
    local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
    current outbound spi: 0x0(0)

  inbound esp sas:

--More--
```

→I Verified that no packets were yet encrypted (all counters at 0).

2. Created interesting traffic by pinging PC-C from PC-A.



```
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

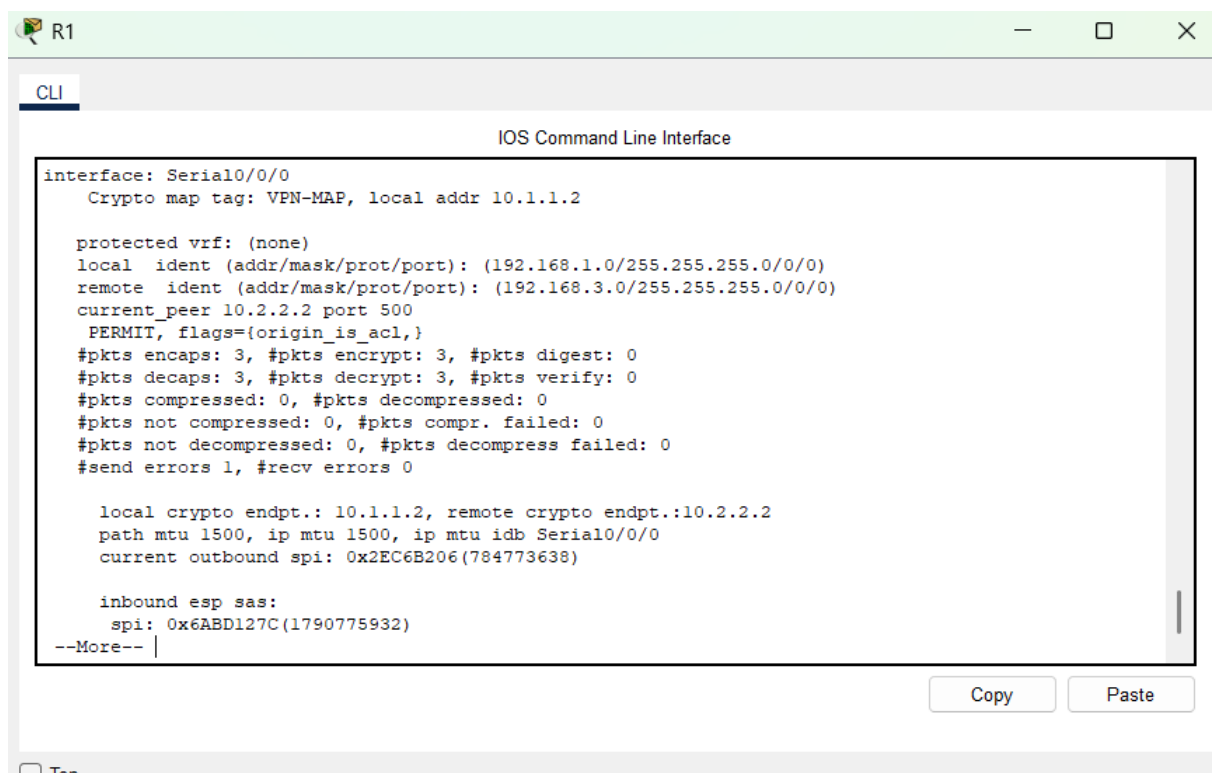
Request timed out.
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126
Reply from 192.168.3.3: bytes=32 time=16ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 16ms, Average = 6ms

C:\>
```

3. Re-checked IPsec status:

R1# show crypto ipsec sa



```
CLI
IOS Command Line Interface

interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

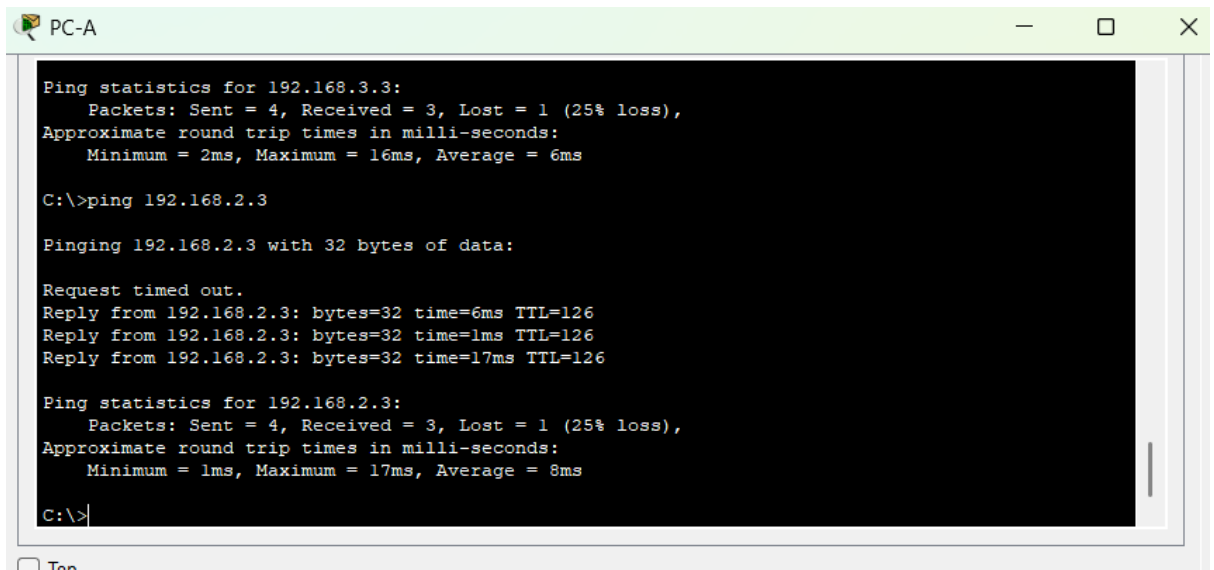
  protected vrf: (none)
  local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
    #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

    local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
    current outbound spi: 0x2EC6B206(784773638)

    inbound esp sas:
      spi: 0x6ABD127C(1790775932)
    --More--
```

→I hence Confirmed that packet counters incremented (VPN tunnel active).

4. Created uninteresting traffic by pinging PC-B from PC-A to ensure no VPN was triggered.



```
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 16ms, Average = 6ms

C:\>ping 192.168.2.3

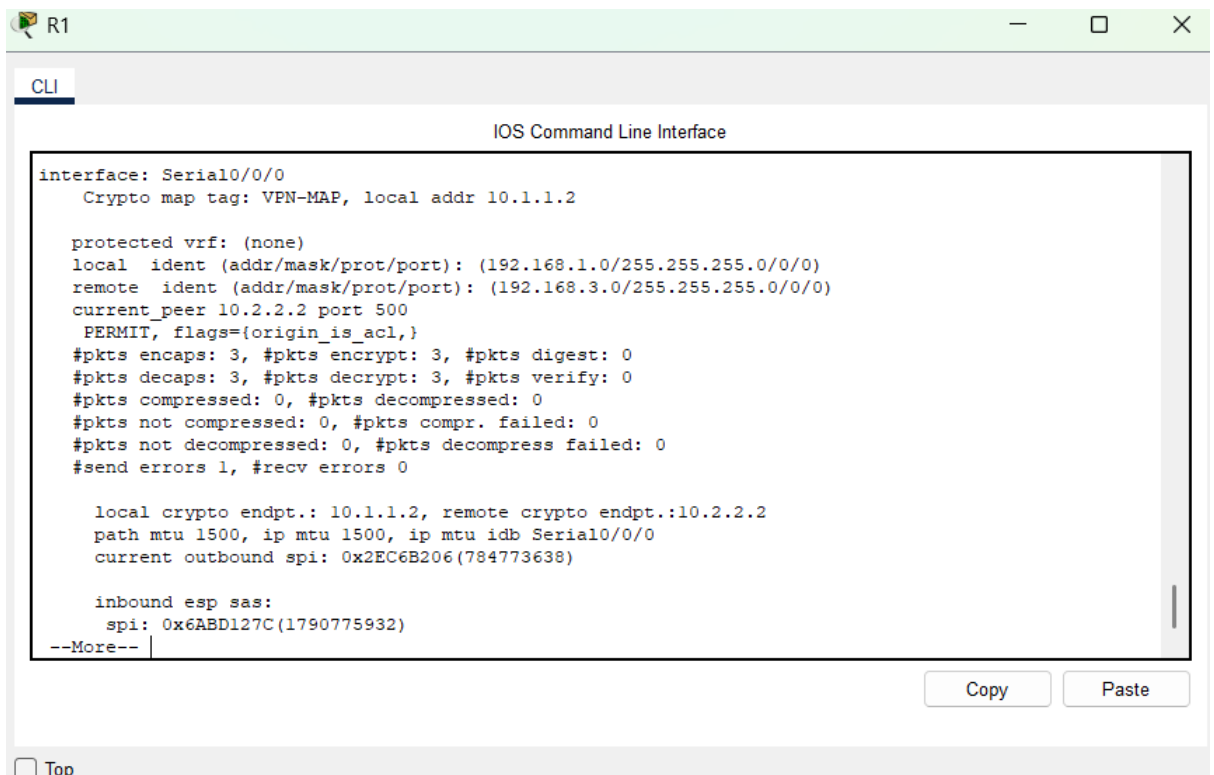
Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=6ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=17ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 17ms, Average = 8ms

C:\>
```

The ping was successful, however there was no encryption as shown below:



```
CLI
IOS Command Line Interface

interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 10.1.1.2

protected vrf: (none)
local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
current_peer 10.2.2.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
#pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
current outbound spi: 0x2EC6B206(784773638)

inbound esp sas:
  spi: 0x6ABD127C(1790775932)
--More--
```

The number of encrypted packets is still as it was on the previous ping no additions meaning no encryption was done on the uninteresting traffic.

In summary, in this section I used the following commands, and these were their purpose:

- `show crypto ipsec sa` shows IPsec Security Associations. You'll use this to see if packets are being encrypted and decrypted, which confirms the tunnel is working.

- `ping 192.168.3.x` from PC-A initiates the interesting traffic to trigger the VPN tunnel.
- After the ping, running `show crypto ipsec sa` again shows incremented packet counters, which proves that traffic is now going through the encrypted tunnel.
- `show crypto isakmp sa` checks whether the IKE Phase 1 negotiation was successful.
- `show crypto map` shows you the crypto map configuration applied to the interfaces

Conclusion

By completing this lab, I gained practical experience in configuring and verifying an IPsec VPN tunnel between two routers. I learned how to apply security policies and how VPN tunnels are activated based on interesting traffic. The activity demonstrated how secure communication can be established across a public network while ensuring data integrity and confidentiality. This exercise solidified my understanding of VPN principles and router-based security configuration using Cisco technologies